

Chapter 2: Taylor Polynomials, Little-o and Big-O Notation

National High School of Mathematics

Contents

1	Taylor Polynomial	1
2	Big-O and Little-o Notation (Bachmann–Landau Notation)	4
2.1	Definition: Little-o	4
2.2	Definition: Big-O	4
2.3	Equivalence Relation ($f \sim g$)	5
3	Limited Development (LD)	5
3.1	Formal Definition of LD	5
3.2	LD of Elementary Functions at $x_0 = 0$ to Order n	6
3.3	LD at a general point x_0	7
4	LD Operation Rules	7
5	Generalized and Asymptotic LD	8
5.1	Application: Limits and Tangent Position	8
5.2	LD at $+\infty$ (Asymptotic LD)	8
A	Extra Sources	10

1. Taylor Polynomial

Let $x_0 \in [a, b]$ and let f be a function defined on $[a, b]$. Suppose that $f(x_0), f'(x_0), f''(x_0), \dots, f^{(n)}(x_0)$ exist for some $n \in \mathbb{N}$. Then the polynomial:

$$P_n(x) = f(x_0) + f'(x_0)(x - x_0) + \dots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

is called the **n^{th} -order Taylor polynomial** associated with f , based at x_0 .

The Taylor polynomial of a function f at a point x_0 is given by:

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

It is natural to think that $P_n(x)$ is a good approximation of f in a neighborhood of x_0 .

Let

$$R_n(x) = f(x) - P_n(x)$$

be the **difference (error)** or the **remainder**. Thus, $P_n(x)$ is called the **differentiable approximation** of f based on x_0 .

Let $f(x) = \sin(x)$ at $x_0 = 1$. Then: $P_1(x) = \sin(1) + \cos(1)(x - 1)$. The remainders might be conceptualized, but the examples given in the original text are somewhat unclear. For instance, for $x_0 = 0$, $P_1(x) = x$, $P_3(x) = x - \frac{x^3}{6}$.

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be two functions such that:

- $f^{(l)}$ are continuous on $[a, b]$ for $l = 0, 1, \dots, n$,
- $f^{(n+1)}$ exists on $]a, b[$,
- g is continuous on $[a, b]$ and g' exists on $]a, b[$,
- $g'(x) \neq 0$ for all $x \in]a, b[$.

Let $x_0 \in [a, b]$. Then, for all $x \in [a, b]$ with $x \neq x_0$, there exists c between x_0 and x such that:

$$f(x) = P_n(x) + R_n(x)$$

where the remainder $R_n(x)$ is given by:

$$R_n(x) = \frac{f^{(n+1)}(c)}{n!g'(c)}(x - c)^n(g(x) - g(x_0))$$

• Proof Sketch

Let $\phi : [x_0, x] \rightarrow \mathbb{R}$ be defined as $\phi(t) = f(x) - \sum_{k=0}^n \frac{f^{(k)}(t)}{k!} (x-t)^k$. We find that $\phi(x) = 0$ and $\phi(x_0) = R_n(x)$. The derivative is $\phi'(t) = -\frac{f^{(n+1)}(t)}{n!} (x-t)^n$. Using the Extended Mean Value Theorem (Cauchy's MVT) for $\phi(t)$ and $g(t)$ on the interval $[x_0, x]$, there exists $c \in (x_0, x)$ such that:

$$\frac{\phi(x) - \phi(x_0)}{g(x) - g(x_0)} = \frac{\phi'(c)}{g'(c)} \implies \frac{0 - R_n(x)}{g(x) - g(x_0)} = \frac{-\frac{f^{(n+1)}(c)}{n!} (x-c)^n}{g'(c)}$$

Rearranging gives the desired formula for $R_n(x)$. (See also: [Real Analysis 47 — Proof of Taylor's Theorem](#))

Let $f : [a, b] \rightarrow \mathbb{R}$, $f \in C^n[a, b]$, and suppose $f^{(n+1)}$ exists on $]a, b[$. Fix $x_0 \in [a, b]$. Then for any $x \in [a, b]$, there exists c between x_0 and x such that:

$$f(x) = P_n(x) + R_n(x),$$

where

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x-x_0)^{n+1}.$$

• Proof

This is a special case of the generalized remainder with $g(t) = (x-t)^{n+1}$. (See also: [Lagrange Remainder Theorem \(Short and EASY\)](#))

Let $f(x) = \ln(1+x)$, so $f(1) = \ln(2)$. We use the expansion about $x_0 = 0$. The derivatives are $f^{(k)}(x) = (-1)^{k-1} (k-1)! (1+x)^{-k}$. So $f^{(k)}(0) = (-1)^{k-1} (k-1)!$. The Taylor expansion of $\ln(1+x)$ about $x_0 = 0$ is:

$$\ln(1+x) = \sum_{k=1}^n \frac{(-1)^{k-1}}{k} x^k + R_n(x).$$

For $x = 1$, we want to find n such that $|R_n(1)| \leq \frac{1}{100}$. The Lagrange remainder is:

$$|R_n(1)| = \left| \frac{f^{(n+1)}(c)}{(n+1)!} (1-0)^{n+1} \right| = \left| \frac{(-1)^n n! (1+c)^{-(n+1)}}{(n+1)!} \right| = \frac{1}{(n+1)(1+c)^{n+1}},$$

for some $c \in (0, 1)$. Since $c > 0$, we have $(1+c)^{n+1} > 1$. Thus, $|R_n(1)| < \frac{1}{n+1}$. We need $\frac{1}{n+1} \leq \frac{1}{100}$, so $n+1 \geq 100$, which means $n \geq 99$.

Let $f : [a, b] \rightarrow \mathbb{R}$, $f \in C^n([a, b])$, and suppose $f^{(n+1)}$ exists on $]a, b[$. Fix $x_0 \in [a, b]$. Then there exists c between x_0 and x such that the remainder is:

$$R_n(x) = \frac{f^{(n+1)}(c)}{n!} (x - c)^n (x - x_0).$$

This can also be written with $\theta \in (0, 1)$ and $c = x_0 + \theta(x - x_0)$ as:

$$R_n(x) = \frac{f^{(n+1)}(x_0 + \theta(x - x_0))}{n!} (x - x_0)^{n+1} (1 - \theta)^n.$$

Let $n \in \mathbb{N}$ and $a_0, a_1, \dots, a_n \in \mathbb{R}$. The polynomial $P(x) = a_0 + a_1x + \dots + a_nx^n$ can be rewritten as $P(x) = b_0 + b_1(x - x_0) + \dots + b_n(x - x_0)^n$, where $b_0, \dots, b_n \in \mathbb{R}$.

• Proof

By induction, or by noting that the coefficients are given by the derivatives of $P(x)$ at x_0 :

$$b_0 = P(x_0), \quad b_1 = P'(x_0), \quad \dots, \quad b_k = \frac{P^{(k)}(x_0)}{k!}.$$

Let $x_0 \in [a, b]$. Suppose $f : [a, b] \rightarrow \mathbb{R}$ is n -times differentiable at x_0 . Then there exists a function $\epsilon : [a, b] \rightarrow \mathbb{R}$ with $\lim_{x \rightarrow x_0} \epsilon(x) = 0$ such that:

$$f(x) = f(x_0) + \sum_{k=1}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + (x - x_0)^n \epsilon(x).$$

• Proof

Define $\epsilon(x) = \frac{f(x) - P_n(x)}{(x - x_0)^n}$ for $x \neq x_0$ and $\epsilon(x_0) = 0$. We need to show $\lim_{x \rightarrow x_0} \epsilon(x) = 0$. This follows from applying L'Hôpital's rule n times to the fraction, using the definition of the Taylor polynomial coefficients.

When the Peano remainder theorem is applied at $x_0 = 0$, it is often called Young's theorem:

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} x^k + x^n \epsilon(x), \quad \text{where } \lim_{x \rightarrow 0} \epsilon(x) = 0.$$

The term $x^n \epsilon(x)$ is also written as $o(x^n)$.

Let $f : [a, b] \rightarrow \mathbb{R}$ be a function such that $f \in C^{n+1}([a, b])$. Fix $x_0 \in [a, b]$. Then for

any $x \in [a, b]$, we have:

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + \frac{1}{n!} \int_{x_0}^x (x - t)^n f^{(n+1)}(t) dt.$$

• **Proof**

By induction on n , using integration by parts on the integral remainder term.

2. Big-O and Little-o Notation (Bachmann–Landau Notation)

2.1 Definition: Little-o

Let f and g be two functions. We say f is negligible compared to g as $x \rightarrow x_0$, written $f = o(g)$ as $x \rightarrow x_0$, if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all x with $0 < |x - x_0| < \delta$, we have $|f(x)| < \varepsilon|g(x)|$. (See also: [Little o](#))

If g does not vanish in a neighborhood of x_0 (except possibly at x_0), then:

$$f = o(g) \text{ as } (x \rightarrow x_0) \quad \Leftrightarrow \quad \lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 0.$$

If $g(x) = 1$, then:

$$f = o(1) \text{ as } (x \rightarrow x_0) \quad \Leftrightarrow \quad \lim_{x \rightarrow x_0} f(x) = 0.$$

2.2 Definition: Big-O

We say f is dominated by g as $x \rightarrow x_0$, written $f = O(g)$ as $x \rightarrow x_0$, if there exist constants $C > 0$ and $\delta > 0$ such that for all x with $0 < |x - x_0| < \delta$, we have $|f(x)| \leq C|g(x)|$. This means the ratio $\frac{f(x)}{g(x)}$ is bounded in a neighborhood of x_0 .

2.3 Equivalence Relation ($f \sim g$)

For x_0 an accumulation point of D , and $f, g : D \rightarrow \mathbb{R}$. We say f is equivalent to g as $x \rightarrow x_0$, written $f \sim g$ as $x \rightarrow x_0$, if

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 1.$$

This is equivalent to $f(x) - g(x) = o(g(x))$ as $x \rightarrow x_0$.

Let $y = x \ln a$. As $x \rightarrow 0$, $y \rightarrow 0$.

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x \ln a} = \lim_{y \rightarrow 0} \frac{e^y - 1}{y} = 1.$$

Thus, $a^x - 1 \sim x \ln a$ as $x \rightarrow 0$. This can be written as $a^x - 1 = x \ln a + o(x \ln a)$, or more simply $a^x - 1 = x \ln a + o(x)$.

3. Limited Development (LD)

The **order** of a Limited Development (LD) is determined by its **remainder**, not the degree of the regular part. **Key Properties:**

- If f admits an LD of order 0, it is **continuous** at x_0 .
- If f admits an LD of order 1, it is **differentiable** at x_0 .
- If f admits an LD of order ≥ 2 , we cannot conclude higher smoothness from the existence alone.

3.1 Formal Definition of LD

Let I be an open interval and $f : I \rightarrow \mathbb{R}$. For $x_0 \in I$ and $n \in \mathbb{N}$, we say f admits a **Limited Development (LD)** at x_0 of order n if there exist constants $c_0, c_1, \dots, c_n \in \mathbb{R}$ and a function $\varepsilon : I \rightarrow \mathbb{R}$ with $\lim_{x \rightarrow x_0} \varepsilon(x) = 0$ such that for all $x \in I$:

$$f(x) = c_0 + c_1(x - x_0) + \dots + c_n(x - x_0)^n + (x - x_0)^n \varepsilon(x)$$

This is also written as:

$$f(x) = \sum_{k=0}^n c_k (x - x_0)^k + o((x - x_0)^n).$$

The expression is composed of a **Regular Part** (the polynomial) and a **Remainder**.

If f admits a limited development (LD) at x_0 to order n , then the polynomial part is unique. If f is n -times differentiable at x_0 , then the coefficients are given by the Taylor coefficients, $c_k = \frac{f^{(k)}(x_0)}{k!}$.

• Proof Sketch of Uniqueness

By contradiction, suppose f admits two different LDs, $P_n(x) + o((x - x_0)^n)$ and $Q_n(x) + o((x - x_0)^n)$. Then $P_n(x) - Q_n(x) = o((x - x_0)^n)$. Let $R(x) = P_n(x) - Q_n(x) = \sum_{k=0}^n (c_k - d_k)(x - x_0)^k$. If not all coefficients are zero, let j be the smallest index such that $c_j \neq d_j$. Then $R(x) \sim (c_j - d_j)(x - x_0)^j$ as $x \rightarrow x_0$. But we also have $R(x) = o((x - x_0)^n)$, which implies $\lim_{x \rightarrow x_0} \frac{R(x)}{(x - x_0)^j} = 0$ since $j \leq n$. This forces $c_j - d_j = 0$, a contradiction.

If f is an even (resp. odd) function, then the polynomial part of its LD at $x_0 = 0$ contains only even (resp. odd) powers of x .

3.2 LD of Elementary Functions at $x_0 = 0$ to Order n

1. $e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + o(x^n)$
2. $\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^p \frac{x^{2p+1}}{(2p+1)!} + o(x^{2p+2})$
3. $\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^p \frac{x^{2p}}{(2p)!} + o(x^{2p+1})$
4. $\sinh(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + \frac{x^{2p+1}}{(2p+1)!} + o(x^{2p+2})$
5. $\cosh(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + \frac{x^{2p}}{(2p)!} + o(x^{2p+1})$
6. $\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + o(x^n)$
7. $\frac{1}{1+x} = 1 - x + x^2 - \cdots + (-1)^n x^n + o(x^n)$

$$8. \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n-1} \frac{x^n}{n} + o(x^n)$$

$$9. (1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \cdots + \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{n!} x^n + o(x^n)$$

3.3 LD at a general point x_0

To find the LD of $f(x)$ at x_0 , let $h = x - x_0$ and find the LD of $g(h) = f(x_0 + h)$ at $h = 0$.

Let $h = x - \pi$.

$$e^x = e^{\pi+h} = e^\pi e^h = e^\pi \left(1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + o(h^3) \right)$$

$$e^x = e^\pi \left(1 + (x - \pi) + \frac{(x - \pi)^2}{2} + \frac{(x - \pi)^3}{6} \right) + o((x - \pi)^3)$$

4. LD Operation Rules

Let $f(x) = P_n(x) + o((x - x_0)^n)$ and $g(x) = Q_n(x) + o((x - x_0)^n)$.

- **Sum/Difference:** $(f \pm g)(x) = (P_n(x) \pm Q_n(x)) + o((x - x_0)^n)$.
- **Product:** $(f \cdot g)(x) = \text{trunc}_n(P_n(x) \cdot Q_n(x)) + o((x - x_0)^n)$, where trunc_n means keeping terms of degree up to n .
- **Composition:** If $f(x_0) = 0$, then $(g \circ f)(x) = \text{trunc}_n(Q_n(P_n(x))) + o((x - x_0)^n)$.
- **Division:** If $Q_n(x_0) \neq 0$, perform polynomial long division of $P_n(x)$ by $Q_n(x)$ up to order n .
- **Integration:** If f has an LD of order n at x_0 , then $F(x) = \int_{x_0}^x f(t)dt$ has an LD of order $n + 1$ at x_0 , obtained by integrating the LD of $f(x)$ term by term.

$$\tan(x) = \frac{\sin(x)}{\cos(x)} = \frac{x - \frac{x^3}{6} + o(x^4)}{1 - \frac{x^2}{2} + o(x^3)}$$

Let $u = -\frac{x^2}{2}$. We have $\frac{1}{1+u} \approx 1 - u$.

$$\begin{aligned}\tan(x) &= \left(x - \frac{x^3}{6}\right) \left(1 - \left(-\frac{x^2}{2}\right) + o(x^3)\right) + o(x^3) \\ &= \left(x - \frac{x^3}{6}\right) \left(1 + \frac{x^2}{2}\right) + o(x^3) \\ &= x + \frac{x^3}{2} - \frac{x^3}{6} + o(x^3) = x + \frac{x^3}{3} + o(x^3)\end{aligned}$$

5. Generalized and Asymptotic LD

5.1 Application: Limits and Tangent Position

Let the LD of f at x_0 be $f(x) = a_0 + a_1(x - x_0) + a_k(x - x_0)^k + o((x - x_0)^k)$, where a_k is the first non-zero coefficient after a_1 .

- The tangent line to the curve $y = f(x)$ at x_0 is $T(x) = a_0 + a_1(x - x_0)$.
- The position of the curve relative to the tangent is given by the sign of $f(x) - T(x) \approx a_k(x - x_0)^k$.
- If k is odd, the curve crosses the tangent at x_0 , which is an **inflection point**.
- If k is even, the curve stays on one side of the tangent near x_0 , which is a local extremum if $a_1 = 0$.

5.2 LD at $+\infty$ (Asymptotic LD)

A function $f(x)$ admits an LD at $+\infty$ if $g(t) = f(1/t)$ admits an LD at $t = 0^+$. This means $f(x)$ can be written as:

$$f(x) = a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} + o\left(\frac{1}{x^n}\right).$$

This form is used to find horizontal asymptotes and the position of the curve relative to them. The line $y = a_0$ is a horizontal asymptote.

If $f(x)$ has an oblique asymptote $y = ax + b$, we can find it by finding the asymptotic LD of $\frac{f(x)}{x}$. If $\frac{f(x)}{x} = a + \frac{b}{x} + o\left(\frac{1}{x}\right)$, then $f(x) = ax + b + o(1)$, and $y = ax + b$ is the asymptote.

Let $t = 1/x$. As $x \rightarrow \infty$, $t \rightarrow 0$.

$$\begin{aligned} f(x) &= \frac{1}{t} \ln(2+t) = \frac{1}{t} \ln \left(2 \left(1 + \frac{t}{2} \right) \right) = \frac{1}{t} \left(\ln 2 + \ln \left(1 + \frac{t}{2} \right) \right) \\ &= \frac{1}{t} \left(\ln 2 + \left(\frac{t}{2} - \frac{1}{2} \left(\frac{t}{2} \right)^2 + o(t^2) \right) \right) \\ &= \frac{\ln 2}{t} + \frac{1}{2} - \frac{t}{8} + o(t) \\ f(x) &= x \ln 2 + \frac{1}{2} - \frac{1}{8x} + o\left(\frac{1}{x}\right) \end{aligned}$$

The oblique asymptote is $y = (\ln 2)x + \frac{1}{2}$.

A. Extra Sources

- **Analyse 2** — les développements limites et les formules de Taylor —
(Playlist in French/Arabic)

<https://www.youtube.com/playlist?list=PLgKSr97JkHD7ef01ULNNmIaNsic8XHpkt>

- **Taylor series** — Chapter 11, Essence of calculus (3Blue1Brown)

<https://www.youtube.com/watch?v=3d6DsjiBzJ4>

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